



DEALER BEST PRACTICE SERIES

OHT, LMT Rear Suspension
Test Bench

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
---------------------------	-------------	---------------------------------	-------------------------------	--------------------

Rear Suspension Test Bench	0
1.0 Introduction	1
2.0 Best Practice Description	1
3.0 Implementation Steps.....	4
4.0 Benefits	4
5.0 Resources required	4
6.0 Supporting Attachments / References	4
7.0 Related Best Practices	4
8.0 Acknowledgements	5

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

This best practice is dealing with Component life management. The current process was inefficient and resulted in high levels of variability. By implementing this Best Practice, there will be fewer repairs after failure and longer component life. This is important to both the dealer and customer, as it will improve customer satisfaction and profitability through reduced down time.

The key benefit from this Best Practice is improved quality. This Best Practice is not available for front suspension testing.

2.0 Best Practice Description

The current process at our CRC is that rear suspensions are not being tested for leaks after rebuild. Components are stripped cleaned and inspected and then stored until approval is received for assembly.

We found that we required a test bench in order to ensure that the cylinders would hold pressure, were rebuilt correctly, and would meet projected life.

BEFORE



THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT CATERPILLAR'S WRITTEN PERMISSION.			
Rear Suspension Test Bench	DATE 6/9/2017	CHG NO 00	NUMBER 1108-4.03-1257



Requirement was to design and fabricate a multi-purpose test bench for various OHT suspensions. This would also have the capability to test the suspension cylinders under pressure to ensure they have no internal leaks.

AFTER

TOOLING REQUIRED



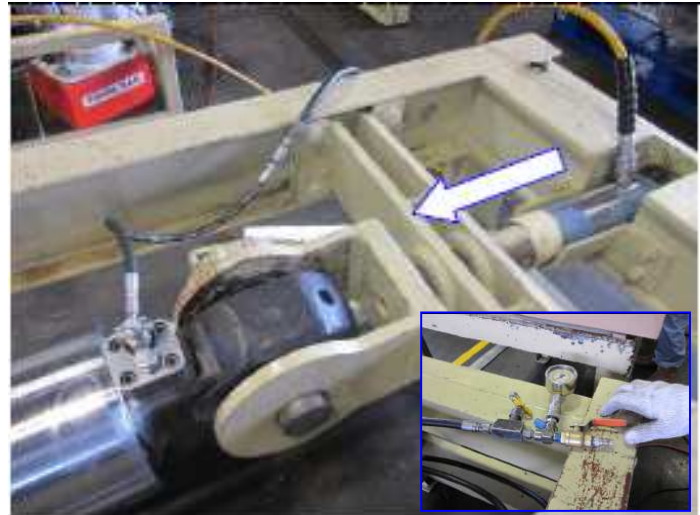
THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT CATERPILLAR'S WRITTEN PERMISSION.

Rear Suspension Test Bench

DATE
6/9/2017

CHG
NO
00

NUMBER
1108-4.03-1257

**TESTING REPORT**

773D BDA Rear Suspension GP
116-2837

The Customer Services Company

Test Pressure

- Fill Hydraulic Oil into the Housing Assembly (with Oil SAE 10W) → 12 liter
- Add Hydraulic Oil Additive
- Connect hose assembly from nitrogen supply tank to the suspension gp. On the supply tank, adjust the regulated pressure to 150 Psi.

Static			
	Fill the column below	Mechanic	Lead hand
Increase Pressure	... 100 ... (Psi)	2330	OK
Check for leaks	Yes / No - Leaks	✓	OK
Did the suspension hold pressure	Yes / No	✓	OK
Time	... 60 ... (Sec.)		OK

Dynamic			
	Fill the column below	Mechanic	Lead Hand
Move the Rod Extend & Retract for five (5) stroke		2330	OK
Check for leaks	Yes / No - Leaks	✓	OK
Pressure Gauge	... 100 ... (Psi)		OK
Time	... 60 (Sec/stroke)		OK

Date: 9/9/11

Mechanic Name: _____
S/N: 2330

Foreman Name: _____
S/N: 7437

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT CATERPILLAR'S WRITTEN PERMISSION.

Rear Suspension Test Bench

DATE
6/9/2017CHG
NO
00NUMBER
1108-4.03-1257

Editor's note: Trakindo is currently using the bench for all rear suspension cylinders they produce (773, 775, 777, 785, 789) and has indicated it is capable to test 793 cylinders.

3.0 Implementation Steps

- Determine area in shop to be designated for the cylinder testing bench
- Assure the area has adequate space and access for equipment to move the component
- Confirm the layout for equipment and make changes to the shop to install
- Promote change feedback as new operation problems may surface with the procedure
- Train employees to use the new procedures/equipment
- Review process/equipment performance once establish
- Make any adjustment(s) as needed
- Drawings are attached in this best practice

4.0 Benefits

Dealer benefits observed:

- Saved time
- Improved quality
- Improved technician safety

5.0 Resources required

Facility-

- Machine shop or outside contractor to build test bench
- Oil used on test will be charged to the customer under miscellaneous

6.0 Supporting Attachments / References

CAD Drawing Rear Suspension Test Bench

7.0 Related Best Practices

-

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT CATERPILLAR'S WRITTEN PERMISSION.

Rear Suspension Test Bench	DATE 6/9/2017	CHG NO 00	NUMBER 1108-4.03-1257
----------------------------	------------------	-----------------	--------------------------

8.0 Acknowledgements

This Best Practice was written by:

- David J. Wilde
- Corporate Best Practice
- Product Support Standard and Development
- PT. Trakindo Utama - Head Office
- Email: djwilde@trakindo.co.id
- Office: 62217822373 ext: 1120

Drawing by :

- Pramudya Ananda
- Specialist Facility Development
- Product Support Standard & Development
- PT. Trakindo Utama - Head Office
- Email: pramudya.ananda@trakindo.co.id
- Office: 6221782 2373 ext: 1126

Samarinda CRC staff

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT CATERPILLAR'S WRITTEN PERMISSION.

Rear Suspension Test Bench

DATE
6/9/2017

CHG
NO
00

NUMBER
1108-4.03-1257